biased. When the transistor is kept in darkness, there will be a very few minority charge carrier flow. This makes negligible collector current. When the light is turned '*ON*', and focused at the collector-base junction, the transistor starts conducting and the amplified current starts flowing through the reverse-biased junction. The amount of current flow depends upon the intensity of focused light.

Application

1. They are used in light detection system.
2. They are widely used in high speed reading of computer punching cards.
3. They also used in light operating switches.

6.7. SOLAR CELL

It is widely used in satellites to provide electrical power, when sunlight is incident on a photovoltaic cell. It is converted into electric energy and the energy converter is called solar cell or solar battery. Basically, solar cell is large PN junction diode. It is made up of Silicon and Selenium.


Charles Fritts

History: In the year 1839, photovoltaic effect was discovered by Edmond Becquerel. This process occurs and creates electric voltage, when light is absorbed by the material. In 1883, Charles Fritts created the first solar cell by coating selenium with a thin layer of gold. Silicon solar cells were produced by Bell laboratory during 1953-1956. In 1958, solar cells were used in space.

Construction: Figure 6.7 shows the basic construction of PN junction solar cell. It consists of single semiconductor crystals, which has been doped with both *P* and *N* type impurities, thereby forming a *PN* junction. A glass window is provided at the top of the *P* type layer. The thickness of the *P* layer is small, so that the incident light can easily reach the junction of the diode. A nickel plated ring is provided around the *P* layer, which acts as a positive terminal. A metal contact provided at the bottom of the *N-layer*, acts as negative terminal.

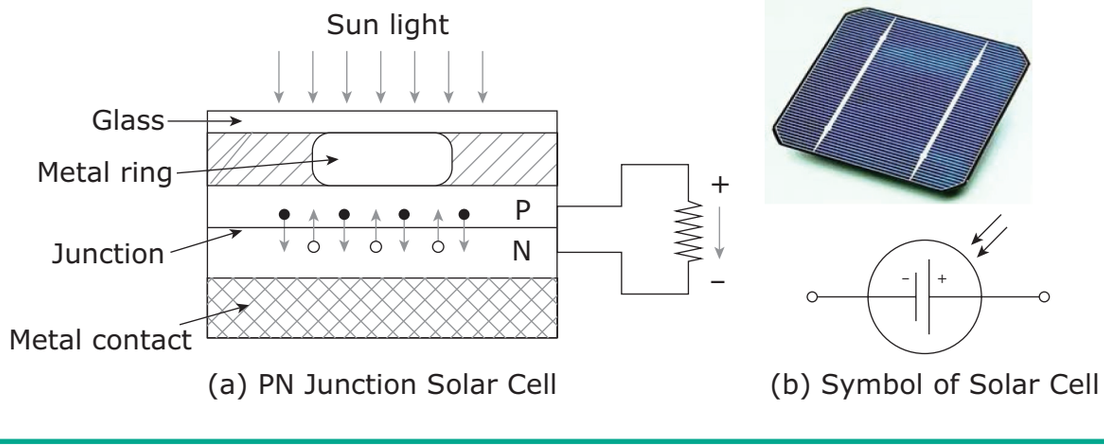


Figure 6.7 **Basic Construction of PN Junction Solar Cell**

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